

Project Initialization and Planning Phase

Date	4 July 2026
Team ID	
Project Title	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

This proposal aims to revolutionize the Human Development Index (HDI) calculation process using Machine Learning, boosting efficiency and analytical accuracy. It tackles manual calculation inefficiencies, promising better data-driven insights and more informed policy-making. Key features include an AI-based prediction model and real-time assessment.

Project Overview	
Objective	To automate and enhance the HDI estimation process by implementing advanced machine learning techniques, ensuring faster and more accurate assessments for policymakers.
Scope	The project comprehensively assesses well-being metrics (GNI, Education, Life Expectancy) and enhances the prediction process, incorporating AI for a more robust and efficient system
Problem Statement	
Description	Addressing inaccuracies and complexities in manual HDI evaluation, which often lacks real-time predictive capabilities.
Impact	Automating these evaluations will result in improved policy intervention planning, reduced analytical errors, and an overall enhancement in development research, contributing to societal progress.
Proposed Solution	
Approach	Employing machine learning techniques (like Regression models) to analyze socio-economic indicators, creating a dynamic and adaptable HDI prediction system.
Key Features	- Implementation of an AI-based multi-dimensional assessment model.

	- Real-time processing for instant prediction results. - Scalability to handle diverse socio-economic datasets.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv